DSC 257R: Unsupervised learning

Solutions to HW 2

1. (a) $\|x\|_1 = 6$
   (b) $\|x\|_2 = \sqrt{14}$
   (c) $\|x\|_\infty = 3$
2. (a) $\|x - x'\|_2 = \sqrt{8}$
   (b) $\|x - x'\|_1 = 4$
   (c) $\|x - x'\|_\infty = 2$
3. *Comparing the ℓ_1, ℓ_2, and ℓ_∞ norms.*
   (a) The point $x = (1, 1, \ldots, 1)$ has $\|x\|_\infty = 1$, but $\|x\|_1 = d$ and $\|x\|_2 = \sqrt{d}$.
   (b) The point $x = (1, 0, \ldots, 0)$ has $\|x\|_2 = 1$ and $\|x\|_\infty = 1$. The point $x = (1, 1, \ldots, 1)/\sqrt{d}$ has $\|x\|_2 = 1$ and $\|x\|_1 = \sqrt{d}$.

   Now let's turn to the inequalities between the three norms. For any $x \in \mathbb{R}^d$, we have

   $$\left(\sum_{i=1}^d |x_i|\right)^2 \;\geq\; \sum_{i=1}^d x_i^2 \;\geq\; \max_i x_i^2$$

   (the first expands to a sum of d^2 non-negative terms which include all the terms in the middle summation). Taking square roots yields $\|x\|_1 \geq \|x\|_2 \geq \|x\|_\infty$.

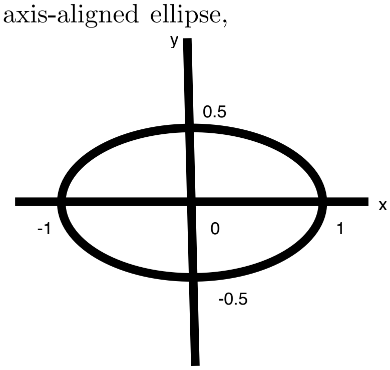
   Likewise,

   $$\sum_{i=1}^d x_i^2 \leq d \max_i x_i^2,$$

   and taking square roots yields $\|x\|_2 \leq \|x\|_\infty \sqrt{d}$. Finally, using Cauchy-Schwarz we have

   $$\|x\|_1 = \sum_{i=1}^d |x_i| \cdot 1 \leq \left(\sum_{i=1}^d x_i^2\right)^{1/2} \left(\sum_{i=1}^d 1^2\right)^{1/2} = \|x\|_2 \sqrt{d}.$$

4. *Weighted ℓ_2 norm.* The unit ball of this norm, $\{(x_1, x_2) : x_1^2 + 4x_2^2 \leq 1\}$, is an axis-aligned ellipse, centered at the origin, that is twice as tall as it is wide.



5. The distance function is **not** a metric. Let's consider the four metric properties:
   - $d(x, y) \geq 0$: satisfied
   - $d(x, y) = 0$ if and only if $x = y$: satisfied
   - $d(x, y) = d(y, x)$: satisfied
   - Triangle inequality: not satisfied. E.g. $d(A, D) > d(A, C) + d(C, D)$.

6. *KL divergence properties.*

   (a) Consider the distributions $p = (1/2, 1/2)$ and $q = (0, 1)$. Then $K(p, q) = \infty$.

   (b) In the same example, $K(q, p) = 1$. Thus the distance function is not symmetric.

7. *Jaccard example.* $J(A, B) = |A \cap B|/|A \cup B| = 3/6 = 1/2$.

8. *Jaccard similarity for text data.* The bigram-sets of the two sentences are

   - {(Napoleon, was), (was, born), (born, in), (in, 1769)}
   - {(Napoleon, was), (was, born), (born, when)}

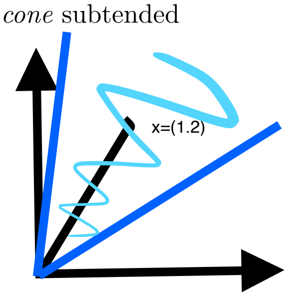
   Thus the bigram-based Jaccard similarity between the sentences is 2/5.

9. *Cosine similarity.*

   (a) $(x \cdot x')/(\|x\|\|x'\|) = 10/14 = 5/7$.

   (b) The cosine similarity between two vectors is zero if and only if the two vectors are orthogonal, that is, at right angles to each other.

   (c) The set of points whose cosine similarity to x is greater than some threshold r is a *cone* subtended at the origin and centered around x.



10. *Loaded dice.*

    (a) The mean is
    $$\mathbb{E}[X] = 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{12} + 4 \cdot \frac{1}{12} + 5 \cdot \frac{1}{12} + 6 \cdot \frac{1}{12} = 2.5$$
    and the median is 2.

    (b) We have
    $$\mathbb{E}[X^2] = 1 \cdot \frac{1}{3} + 4 \cdot \frac{1}{3} + 9 \cdot \frac{1}{12} + 16 \cdot \frac{1}{12} + 25 \cdot \frac{1}{12} + 36 \cdot \frac{1}{12} = \frac{53}{6}$$
    so $\mathrm{var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 53/6 - 25/4 = 31/12$ and $\mathrm{std}(X) = \sqrt{31/12}$.

    (c) The empirical distribution $\widehat{P}$ places mass 1/10 at each observation. Thus
    $$\widehat{P}(1) = 1/5, \quad \widehat{P}(2) = 2/5, \quad \widehat{P}(3) = 0, \quad \widehat{P}(4) = 1/10, \quad \widehat{P}(5) = 1/5, \quad \widehat{P}(6) = 1/10$$

    (d) The empirical distribution has mean
    $$\sum_{j=1}^{6} j\,\widehat{P}(j) = \frac{1}{5} + \frac{4}{5} + 0 + \frac{2}{5} + 1 + \frac{3}{5} = 3,$$
    and median 2. Its variance is
    $$\sum_{j=1}^{6} (j-3)^2\,\widehat{P}(j) = \frac{4}{5} + \frac{2}{5} + 0 + \frac{1}{10} + \frac{4}{5} + \frac{9}{10} = 3$$
    and thus its standard deviation is $\sqrt{3}$.

11. (a) The distribution of H will probably be rather like a bell curve. So its mean will be very similar to its median.

(b) The distribution of C is likely to place non-negligible mass at some values that are much greater than the median. It is likely that the mean will be substantially more than the median.

(c) The distribution of G might be a bell curve or it might be bimodal. But it won't have any outrageously high or low values since grades lie in a very tight range. Thus the median and mean are likely to be close together.

(d) S will likely have a mean that is significantly larger than the median, due to a small fraction of salaries being astronomically high.

12. $\mathbb{E}[Z^2] = \text{var}(Z) + (\mathbb{E}[Z])^2 = \text{std}(Z)^2 + (\mathbb{E}[Z])^2 = 5$.

13. (a) X and Y are not independent. For instance, $\Pr(X = 2) = 0.3$ and $\Pr(Y = 2) = 0.5$ but $\Pr(X = 2, Y = 2) = 0.1 \neq \Pr(X = 2)\Pr(Y = 2)$.

    (b) We have

    - $\Pr(X = 1) = 0.35$, $\Pr(X = 2) = 0.3$, $\Pr(X = 3) = 0.35$. Thus $\mathbb{E}[X] = 2$ and $\text{var}(X) = 0.7$.
    - $\Pr(Y = 1) = 0.3$, $\Pr(Y = 2) = 0.5$, $\Pr(Y = 3) = 0.2$. Thus $\mathbb{E}[Y] = 1.9$ and $\text{var}(Y) = 0.49$.
    - $\Pr(XY = 1) = 0.1$, $\Pr(XY = 2) = 0.3$, $\Pr(XY = 3) = 0.15$, $\Pr(XY = 4) = 0.1$, $\Pr(XY = 6) = 0.3$, $\Pr(XY = 9) = 0.05$. Thus $\mathbb{E}[XY] = 3.8$.
    - $\text{cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$. Thus also $\text{corr}(X, Y) = 0$.

14. *Correlation between linearly related variables.*

    (a) $\mathbb{E}[Y] = a\mathbb{E}[X] + b$ and $\mathbb{E}[XY] = \mathbb{E}[aX^2 + bX] = a\mathbb{E}[X^2] + b\mathbb{E}[X]$. Therefore

    $$\text{cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = (a\mathbb{E}[X^2] + b\mathbb{E}[X]) - \mathbb{E}[X](a\mathbb{E}[X] + b) = a(\mathbb{E}[X^2] - \mathbb{E}[X]^2) = a \cdot \text{var}(X).$$

    (b) Since $\text{std}(Y) = |a| \cdot \text{std}(X)$,

    $$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\text{std}(X)\text{std}(Y)} = \frac{a \cdot \text{var}(X)}{|a| \cdot \text{std}(X)^2} = 1.$$ when a>0, -1 when a<0, not defined when a=0

15. *Do deterministic relationships imply correlation?* Since $\mathbb{E}[X] = 0$, we have $\text{cov}(X, Y) = \mathbb{E}[Xf(X)]$. This is zero if and only if $f(1) = f(-1)$. For instance, $f(-1) = f(1) = 1$ and $f(0) = 10$.